

Tuple protocol for fixed-size span

Abstract

This paper proposes amending fixed-size spans with the tuple protocol, enabling structured binding, integration with `views::elements` and pattern matching once it is approved.

Tony Table

Before	Proposed
<pre>span<int, 3> s{...}; auto & x{s[0]}; auto & y{s[1]}; auto & z{s[2]};</pre>	<pre>span<int, 3> s{...}; auto & [x, y, z]{s};</pre>
<pre>vector<span<int, 3>> ss{...}; auto firsts{ss views::transform(auto s) { return s[0]; }} ranges::to<vector>();</pre> //NOTE: <code>views::transform</code> returns a copy 🙄	<pre>vector<span<int, 3>> ss{...}; auto firsts{ss views::elements<0> ranges::to<vector>()};</pre> //NOTE: <code>views::elements</code> returns by reference 😊
✗ <code>span</code> is not compatible with pattern matching	//interaction with pattern matching proposal P2688R5 <pre>span<double, 2> p{...}; p match { [0, 0] => std::println("at origin"); [let x, 0] => std::println("on x-axis at {}", x); [0, let y] => std::println("on y-axis at {}", y); let [x, y] => std::println("at {}, {}", x, y); };</pre>

Revisions

R0: Initial version

R1: Fixed design issue kindly pointed out by Tomasz Kamiński.

R2: Wording change: not extending **[depr.tuple]** - following the LEWG decision for `complex`.

Motivation

The *tuple-protocol* has been introduced in C++11 and has been supported for `array`, `tuple` and `pair` ever since. With structured bindings (C++17) this protocol was made an integral customisation point for users to tap into a language feature - something that is bound to become even more important with the introduction of pattern matching in a future standard.

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Support for the *tuple-protocol* has been applied to `ranges::subrange` and `complex` in C++20 and C++26 respectively. At the time of writing the only fixed-size library types that do not support structured binding are: `bitset`, `integer_sequence` and `span`.

We can come up with rationales for why the first two in this list do not support it: the former would have to provide proxy-references, something currently not supported by structured binding, the latter is a meta-programming tool primarily used for deduction of its values. For `span` we lack such a clear rationale.

In fact [P1024](#) already proposed this feature - together with several other useful additions and got accepted during the C++20 cycle. After its approval [LWG3212](#) was filled, as the approved design would have resulted in `tuple_element_t<const span<T, 3>>` yielding `const T`. Per [P2116](#) the feature was dropped from C++20.

Design Space

Given the established design of the *tuple-protocol* there is little to discuss, apart from revisiting the issue that previously lead to the removal of this feature: Our design is based on the fact that conceptually `span<T, 1>` is equivalent to `tuple<T &>`, therefore the *tuple-protocol* should be equivalent for these types as well. Which leads us to the following semantics:

- `tuple_size_v<top-level-cv span<cv T, N>> == N`
- `tuple_element_t<I, top-level-cv span<cv T, N>> == cv T &`
- `decltype(get(span<cv T, N>)) == cv T &`

All of which is only valid if `N != dynamic_extent`.

Note that `top-level-cv` (which may either be empty or `const`) is always ignored, staying consistent with the existing language rules for applying `cv`-qualifiers to references. In addition to the above, we adjust the exposition-only *tuple-like* concept to include fixed-size spans, enabling support for adaptors like `views::elements`.

Extending the *tuple-protocol* has one unfortunate side effect: It renders the following previously valid code ambiguous.

```
void f(span<int>) { ... }
void f(tuple<int, int>) { ... }

span<int, 2> s{...};
f(s); // ! ambiguous as either user-defined conversion is valid
```

The same is already true for other *tuple-like* types (see: <https://godbolt.org/z/r65rnPY81>).

```
void f(pair<int, int>) { ... }
void f(tuple<int, int>) { ... }

pair<const int, int> s{...};
f(s); // ✗ ambiguous as either user-defined conversion is valid

array<int, 2> a;
f(s); // ✗ ambiguous as either user-defined conversion is valid

void g(complex<double>) { ... }
void g(pair<float, float>) { ... }

complex<float> c;
g(c); // ✗ ambiguous as either user-defined conversion is valid
```

This type of breakage was already explicitly pointed out in [P2165](#) which introduced the *tuple-like* concept. Therefore we don't consider this issue novel or in any way blocking to this paper.

Impact on the Standard

This proposal is a library extension changing the meaning of `tuple-like<span<T, E>>`, that may result in a breaking change for existing code.

Implementation Experience

The proposed design has been implemented on Godbolt (<https://godbolt.org/z/vrKffnMWr>) and by Tomasz Kamiński at <https://gcc.gnu.org/pipermail/libstdc++/2025-October/063762.html>.

Proposed Wording

Wording is relative to [N5032]. Additions are presented like **this**, removals like **this** and drafting notes like **this**.

[version.syn]

```
#define __cpp_lib_tuple_like 202311LYYYYMML //also in <utility>, <tuple>, <map>, <unordered_map>, <span>
```

[DRAFTING NOTE: Adjust the placeholder value as needed to denote the proposal's date of adoption.]

[tuple.like]

???.?? Concept *tuple-like*

[tuple.like]

```
template<class T>
concept tuple-like = see below; //exposition only
```

1 A type T models and satisfies the exposition-only concept *tuple-like* if `remove_cvref_t<T>` is a specialization of ~~`array`, `complex`, `pair`, `tuple`, or `ranges::subrange`~~.

(1.1) ~~`array`, `complex`, `pair`, `tuple`, `ranges::subrange` or~~

(1.2) ~~`span` and `remove_cvref_t<T>::extent` is not equal to `dynamic_extent`.~~

[views.contiguous]

???.?? Header `<span>` synopsis

[span.syn]

```
// mostly freestanding
namespace std {
...
// [views.span], class template span
...
template<class ElementType, size_t Extent>
constexpr bool ranges::enable_borrowed_range<span<ElementType, Extent>> = true;

// [span.tuple]. tuple interface
template<class T> struct tuple_size;
template<size_t I, class T> struct tuple_element;
template<class ElementType, size_t Extents>
struct tuple_size<span<ElementType, Extents>>;
template<size_t I, class ElementType, size_t Extents>
struct tuple_element<I, span<ElementType, Extents>>;
template<size_t I, class ElementType, size_t Extents>
constexpr ElementType& get(span<ElementType, Extents>) noexcept;

// [span.objectrep], views of object representation
...
}
```

???.?? Class template `span`

[views.span]

...

???.?? Iterator support

[span.iterators]

...

```
constexpr reverse_iterator rend() const noexcept;
```

6 Effects: Equivalent to: `return reverse_iterator(begin());`

???.?? Tuple interface

[span.tuple]

```
template<class ElementType, size_t Extents>
struct tuple_size<span<ElementType, Extents>> : integral_constant<size_t, Extents> {};
```

1	<i>Constraints:</i> <code>Extents != dynamic_extents</code> is true.	
	<code>template<size_t I, class ElementType, size_t Extents></code> <code>struct tuple_element<I, span<ElementType, Extents>> {</code> <code>using type = ElementType&;</code> <code>};</code>	
2	<i>Mandates:</i>	
(2.1)	<code>Extents != dynamic_extents</code> is true, and	
(2.2)	<code>I < Extents</code> is true.	
	<code>template<size_t I, class ElementType, size_t Extents></code> <code>constexpr ElementType& get(span<ElementType, Extents> s) noexcept;</code>	
3	<i>Mandates:</i>	
(3.1)	<code>Extents != dynamic_extents</code> is true, and	
(3.2)	<code>I < Extents</code> is true.	
4	<i>Effects:</i> Equivalent to: <code>return s[I];</code>	
???? Views of object representation		<code>[span.objectrep]</code>

[tuple.helper]

	???? Tuple helper classes	<code>[tuple.helper]</code>
	...	
	<code>template<class T> struct tuple_size<const T>;</code>	
	...	
6	In addition to being available via inclusion of the <code><tuple></code> header, the two templates are available when any of the headers <code><array></code> (<code>[array.syn]</code>), <code><complex></code> (<code>[complex.syn]</code>), <code><ranges></code> (<code>[ranges.syn]</code>), <code></code> (<code>[span.syn]</code>), or <code><utility></code> (<code>[utility.syn]</code>) are included.	
	<code>template<size_t I, class T> struct tuple_element<I, const T>;</code>	
	...	
8	In addition to being available via inclusion of the <code><tuple></code> header, the two templates are available when any of the headers <code><array></code> (<code>[array.syn]</code>), <code><complex></code> (<code>[complex.syn]</code>), <code><ranges></code> (<code>[ranges.syn]</code>), <code></code> (<code>[span.syn]</code>), or <code><utility></code> (<code>[utility.syn]</code>) are included.	

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